

Chapter 3 - Configuring Color and Text with CSS

Overview of Cascading Style Sheets (CSS)

See what is possible with CSS:

- Visit <http://www.csszengarden.com> Style Sheets
- used for years in Desktop Publishing
- apply typographical styles and spacing to printed media

CSS

- provides the functionality of style sheets (and much more) for web developers
- a flexible, cross-platform, standards-based language developed by the W3C.

CSS Advantages

Greater typography and page layout control Style is separate from structure

Styles can be stored in a separate document and associated with the web page

Potentially smaller documents Easier site maintenance

Types of Cascading Style Sheets

Inline Styles Embedded Styles External Styles

Imported Styles

Types of Cascading Style Sheets

- Inline Styles
 - body section
 - HTML style attribute
 - apply only to the specific element
- Embedded Styles
 - head section
 - HTML style element
 - apply to the entire web page document
- External Styles
 - Separate text file with .css file extension
 - Associate with a HTML link element in the head section of a web page
- Imported Styles
 - Similar to External Styles
 - We'll concentrate on the other three types of styles.

CSS Syntax

- Style sheets are composed of “Rules” that describe the styling to be applied.
- Each Rule contains a Selector and a Declaration



Figure 3.2 Using CSS to set the text color to blue

CSS Syntax Sample

Configure a web page to display blue text and yellow background.

```
body { color: blue;  
        background-color: yellow; }
```

This could also be written using hexadecimal color values as shown below.

```
body { color: #0000FF;  
        background-color: #FFFF00; }
```

Common Formatting CSS Properties

- Common CSS Properties, including:
 - background-color
 - color
 - font-family
 - font-size
 - font-style
 - font-weight
 - line-height
 - margin
 - text-align
 - text-decoration
 - width

Using Color on Web Pages

Computer monitors display color as intensities of red, green, and blue light

RGB Color

The values of red, green, and blue vary from 0 to 255.

Hexadecimal numbers (base 16) represent these color values.

Hexadecimal Color Values

- # indicates a hexadecimal value
- Hex value pairs range from 00 to FF
- Three hex value pairs describe an RGB color

#000000 black

#FF0000

red

#00FF00

green

#0000FF

blue

#CCCCCC

grey



Red: #FF0000



Green: #00FF00



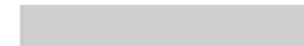
Blue: #0000FF



Black: #000000



White: #FFFFFF



Grey: #CCCCCC

Web Color Palette

#FFFFFF	#FFFFCC	#FFFF99	#FFFF66	#FFFF33	#FFFF00
#FFCCFF	#FFCCCC	#FFCC99	#FFCC66	#FFCC33	#FFCC00
#FF99FF	#FF99CC	#FF9999	#FF9966	#FF9933	#FF9900
#FF66FF	#FF66CC	#FF6699	#FF6666	#FF6633	#FF6600
#FF33FF	#FF33CC	#FF3399	#FF3366	#FF3333	#FF3300
#FF00FF	#FF00CC	#FF0099	#FF0066	#FF0033	#FF0000

Figure 3.5 Partial color chart

Support Web Accessibility Verify Sufficient Contrast

When you choose colors for text and background, sufficient contrast is needed so that the text is easy to read.

Use one of the following online tools to verify contrast:

- <http://webaim.org/resources/contrastchecker>
- http://snook.ca/technical/colour_contrast/colour.html
- <http://juicystudio.com/services/luminositycontrastratio.php>

Configuring Color with Inline CSS

Inline CSS

- Configured in the body of the web page
- Use the style attribute of an HTML tag
- Apply only to the specific element

Example: configure red color text in an <h1> element:

```
<h1 style="color:#ff0000">Heading text is red</h1>
```



Configuring Color with Inline CSS (2)

Example 2: configure the red text in the heading configure a gray background in the heading

Separate style rule declarations with a ; symbol

```
<h1 style="color: #FF0000; background- color: #cccccc">This is displayed as a red heading with gray background</h1>
```

CSS Embedded (Internal) Styles

Configured in the head section of a web page. Use the HTML <style> element

Apply to the entire web page document

Style declarations are contained between the opening and closing <style> tags

Example: Configure a web page with white text on a black background

```
<style>
```

```
body { background-color: #000000; color: #FFFFFF;
```

```
}
```

```
</style>
```

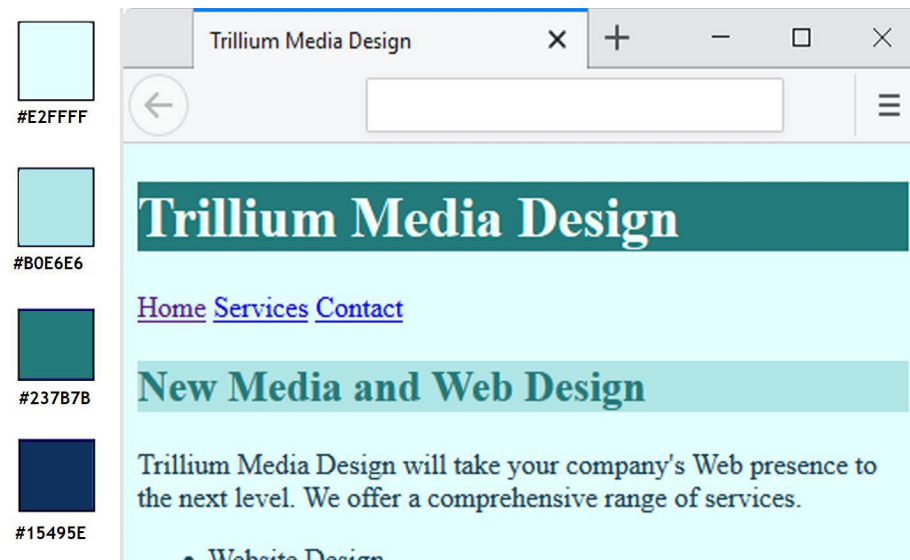


Figure 3.10 The web page after embedded styles are configured

CSS Embedded Styles

- The body selector sets the global style rules for the entire page.
- These global rules are overridden for <h1> and <h2> elements by the h1 and h2 style rules.

<style>

body { background-color: #E2FFFF; color: #15495E;}

**h1 { background-color: #237B7B;
color: #E2FFFF;}**

**h2 { background-color: #B0E6E6;
color: #237B7B;}**

</style>

Configuring Text with CSS

CSS properties for configuring text:

- font-weight
 - Configures the boldness of text
- font-style
 - Configures text to an italic style
- font-size
 - Configures the size of the text
- font-family
 - Configures the font typeface of the text

Text Values	Em Units	Px Units	Pt Units	Percentage
xx-small	.5 em	8 px	6 pt	50%
x-small	.60 em	11 px	8 pt	60%
small	.75 em	13 px	10 pt	75%
medium	1 em	16 px	12 pt	100%
large	1.15 em	18 px	13.5 pt	110%
x-large	1.5 em	24 px	18 pt	150%
xx-large	2 em	30 px	24 pt	200%

The font-size Property

Accessibility Recommendation: Use em or percentage font sizes—these can be easily enlarged in all browsers by users

Font-family Category	Font Typeface
serif	Times New Roman, Georgia, Times
sans-serif	Arial, Verdana, Geneva
monospace	Courier New, Lucida Console
cursive	Brush Script MT, Comic Sans MS
fantasy	Jokerman, Curlz MT

Figure 3.11 Common fonts

The font-family Property

- Not everyone has the same fonts installed in their computer
- Configure a list of fonts and include a generic family name

p { font-family: Arial, Verdana, sans-serif; }

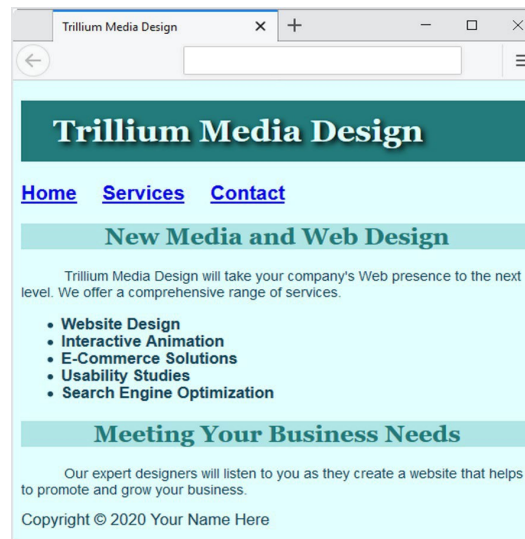


Figure 3.14 CSS configures color and text properties on the web page

Embedded Styles Example

```
<style>
body { background-color: #E2FFFF; color: #15495E; font-family: Arial,
      Verdana, sans-serif; }
h1 { background-color: #237B7B; color: #E2FFFF; line-height: 200%;
      font-family: Georgia, "Times New Roman", serif; text-indent: 1em;
      text-shadow: 3px 3px 5px #000000; }
h2 { background-color: #B0E6E6; color: #237B7B;
      font-family: Georgia, "Times New Roman", serif; text-align: center; }
nav { font-weight: bold; font-size: 1.25em; word-spacing: 1em; }
p { font-size: .90em; text-indent: 3em; } ul { font-weight: bold; }
</style>
```

More CSS TEXT Properties

- line-height
 - Configures the height of the line of text (use the value 200% to appear double-spaced)
- text-align
 - Configures alignment of text within a block display element
- text-indent
 - Configures the indentation of the first line of text
- text-decoration
 - Modifies the appearance of text with an underline, overline, or line-through

More CSS TEXT Properties

- text-transform
 - Configures the capitalization of text
- letter-spacing
 - Configures space between text characters
- word-spacing
 - Configures space between words
- text-shadow
 - Configures a drop shadow on text

CSS Selectors

CSS style rules can be configured for an:

- HTML element selector
- class selector
- id selector
- descendant selector

Using CSS with “class”

Configure with #idname

class Selector

- Apply a CSS rule to a certain “class” of elements on a web page
- Does not associate the style to a specific HTML element

Configure with .classname

<style>

.new { color: #FF0000; font-style: italic;

}

</style>

code CSS to create a class called “new” with red italic text.

Apply the class:

<p class=“new”>This is text is red and in italics</p>

Using CSS with “id”

id Selector

- Apply a CSS rule to ONE element on a web page.

Configure with #idname

<style>

```
#new { color: #FF0000; font-size:2em; font-style: italic;
}
```

</style>

- Code CSS to create an id called “new” with red, large, italic text.

Apply the id:

<p id=“new”>This is text is red, large, and in italics</p>

CSS Descendant Selector

Specify an element within the context of its container (parent) element.
AKA contextual selector

The example configures a green text color only for

<style>

```
#content p { color: #00ff00;
}
```

</style>

p tags located *within* an element assigned to the id named content

Advantage of contextual selectors:

Reduces the number of classes and ids you need to apply in the HTML

span element

Purpose:

- configure a specially formatted area displayed in-line with other elements, such as within a paragraph.

There is no additional empty space above or below a span – it is inline display.

span Element Example

Embedded CSS:

<style>

```
.companyname { font-weight: bold;  
                font-family: Georgia, "Times New Roman", serif;  
                font-size: 1.25em; }
```

</style>

HTML:

```
<p>Your needs are important to us at <span class="companyname">  
>Acme Web Design</span>.  
We will work with you to build your website.</p>
```

Your needs are important to us at **Acme Web Design**. We will work with you to build your website.

External Style Sheets - 1

CSS style rules are contained in a text file separate from the HTML documents.

The External Style Sheet text file:

- extension “.css”
- contains only style rules
- does not contain any HTML tags

External Style Sheets - 2

Multiple web pages can associate with the same external style sheet file.

site.css

```
body {background-color:#E6E6FA;  
       color:#000000;  
       font-family: Arial, sans-  
       serif; font-size:90%; }  
h2 { color: #003366; }  
nav { font-size: 16px;  
      font-weight: bold; }
```

index.html clients.html

about.html

Etc...

link Element

A self-contained tag Placed in the head section

Purpose: associates the external style sheet file with the web page.

Example:

```
<link rel="stylesheet" href="color.css">
```

Using an External Style Sheet

External Style Sheet color.css

```
body { background-color: #0000FF; color: #FFFFFF;
```

```
}
```

To associate the external style sheet called color.css, the HTML code placed in the head section is:

```
<link rel="stylesheet" href="color.css">
```

Centering Page Content with CSS

```
#container { margin-left: auto; margin-right: auto; width: 80%; }
```

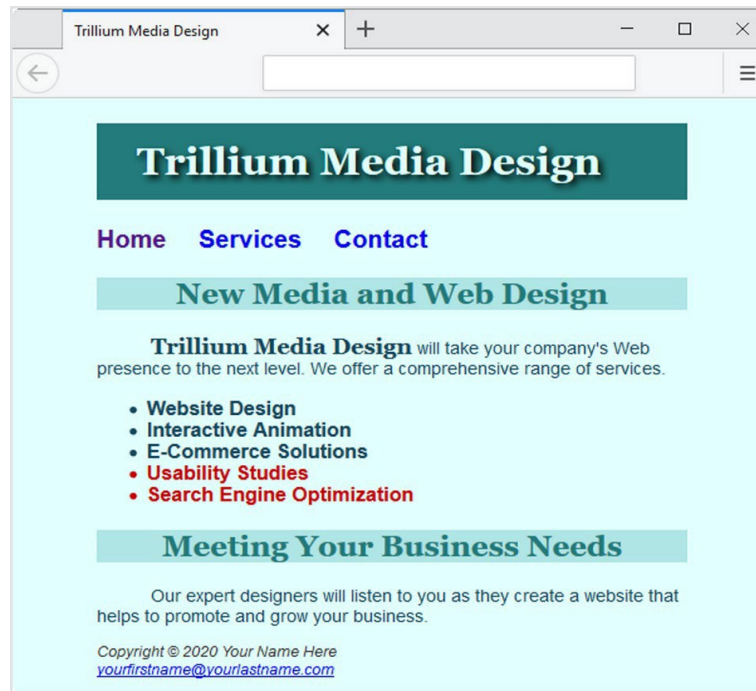



Figure 3.23 The page content is centered within the browser viewport

Summary

This chapter introduced you to Cascading Style Sheet Rules associated with color and text on web pages. You configured inline styles, embedded styles, and external styles. You applied CSS style rules to HTML, class, id, and descendent selectors.